

Báo cáo bài tập Lập trình mạng

Giảng viên hướng dẫn: Lê Bá Vui

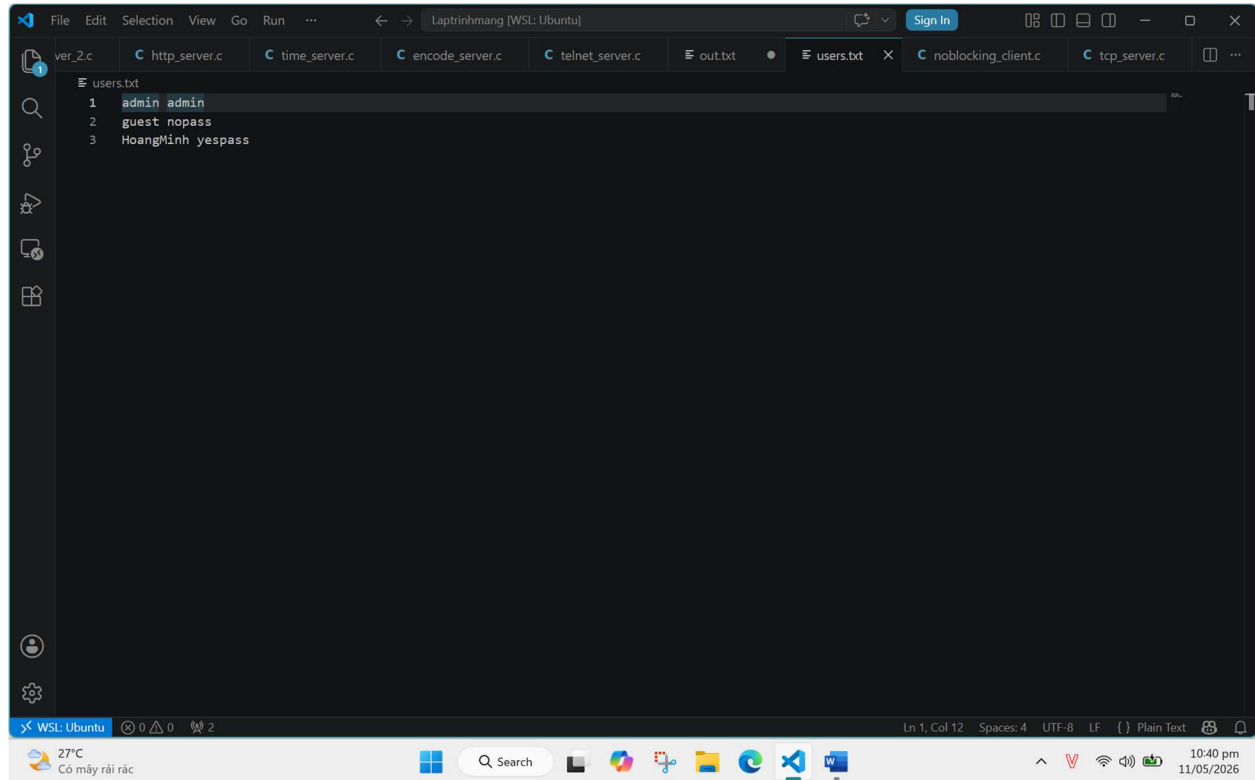
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Mã số sinh viên: 20235377

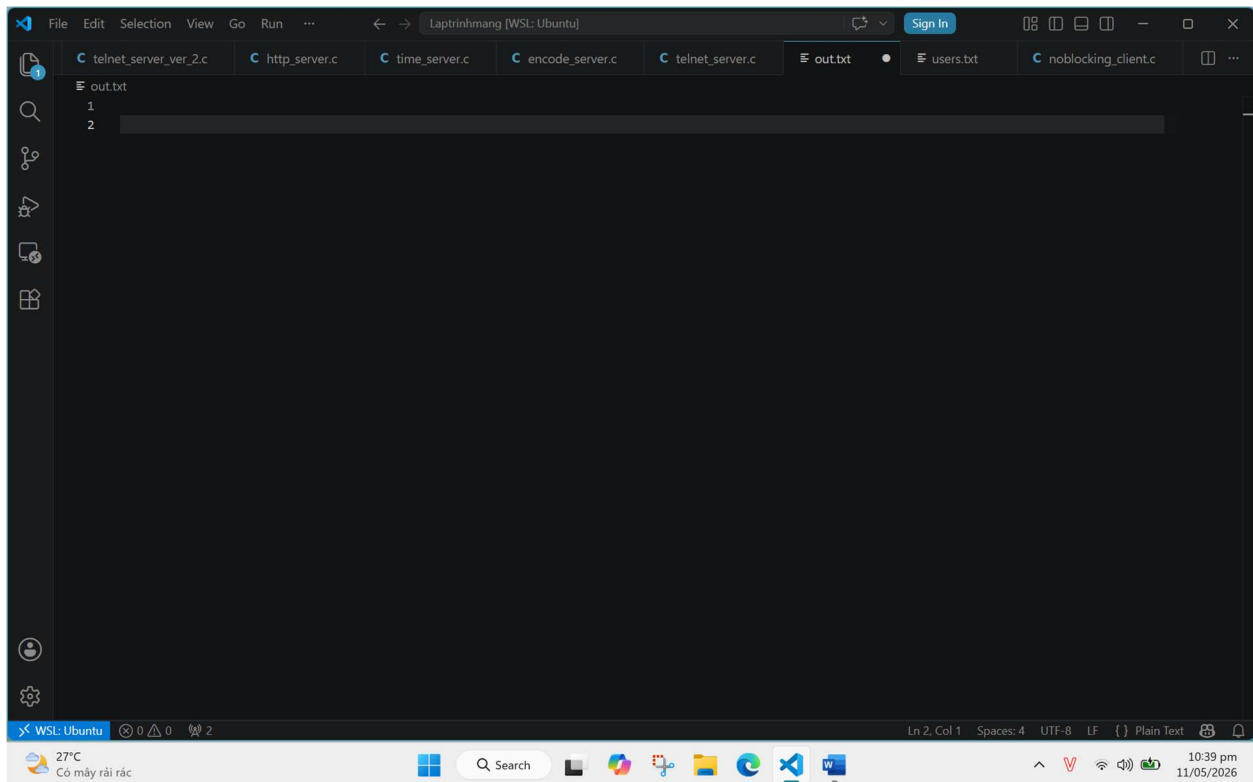
Học phần: IT4060

Mã lớp: 168503

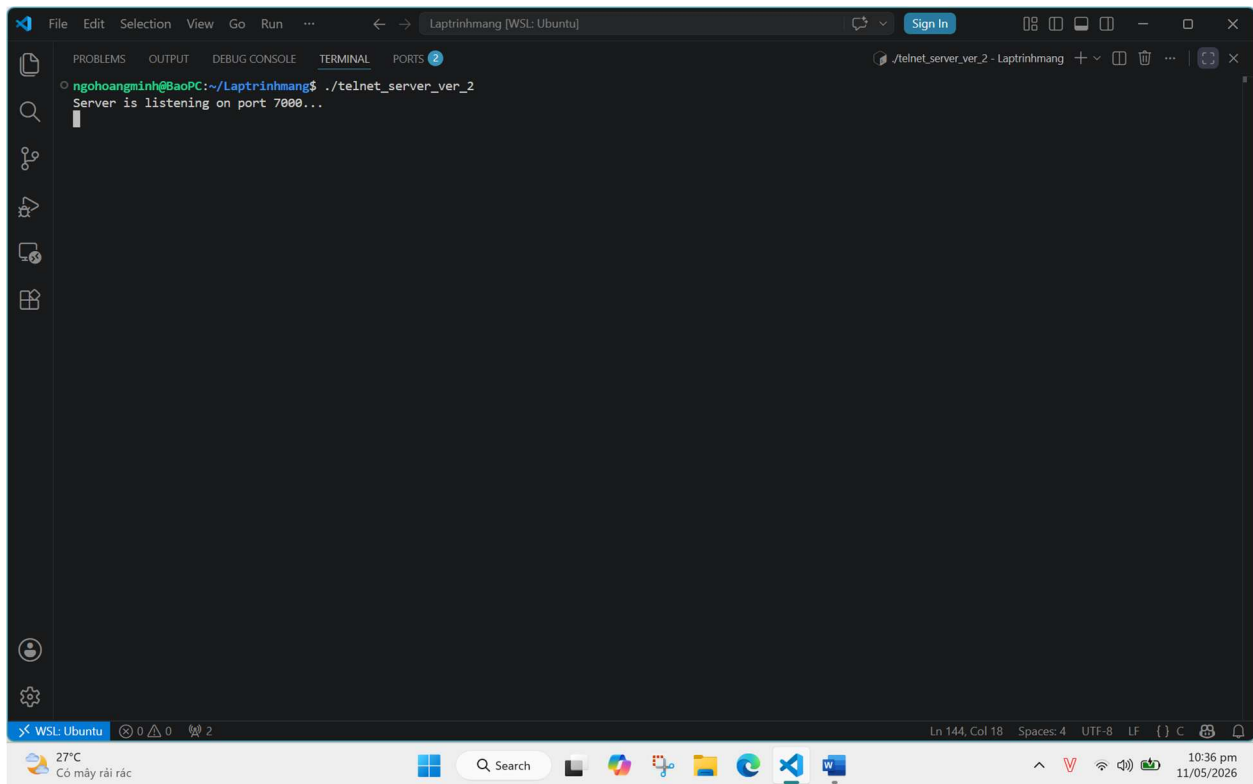
Bài tập 1: Lập trình lại ứng dụng telnet_server sử dụng kỹ thuật multiprocessing.



file users.txt lưu thông tin các tài khoản



file out.txt trước khi chạy chương trình



Khởi chạy server chờ kết nối ở cổng 7000

```
File Edit Selection View Go Run ... Laptrinhmang [WSL: Ubuntu]
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
ngohoangminh@BaoPC:~/Laptrinhmang$ ./telnet_server_ver_2
Server is listening on port 7000...
nc - Laptrinhmang
ngohoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
Nhap user pass (vi du: admin admin):
```

client 1 kết nối tới server

```
File Edit Selection View Go Run ... Laptrinhmang [WSL: Ubuntu]
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
ngohoangminh@BaoPC:~/Laptrinhmang$ ./telnet_server_ver_2
Server is listening on port 7000...
nc - Laptrinhmang
ngohoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
Nhap user pass (vi du: admin admin):
nc - Laptrinhmang
ngohoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
Nhap user pass (vi du: admin admin):
```

client 2 kết nối tới server

```
File Edit Selection View Go Run ... Laptrinhmang [WSL: Ubuntu] Sign In
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
ngoahoangminh@BaoPC:~/Laptrinhmang$ ./telnet_server
Server is listening on port 7000...
ngoahoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
000
Nhap user pass (vi du: admin admin): admin:admin
Sai cu phap! Vui long nhap theo cu phap: user pas
Nhap user pass (vi du: admin admin):
ngoahoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
000
Nhap user pass (vi du: admin admin):
```

client 1 đăng nhập nhưng sai cú pháp

```
File Edit Selection View Go Run ... Laptrinhmang [WSL: Ubuntu] Sign In
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
ngoahoangminh@BaoPC:~/Laptrinhmang$ ./telnet_server
Server is listening on port 7000...
ngoahoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
000
Nhap user pass (vi du: admin admin): admin:admin
Sai cu phap! Vui long nhap theo cu phap: user pas
Nhap user pass (vi du: admin admin):
ngoahoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
000
Nhap user pass (vi du: admin admin): guest matkha
u
Sai tai khoan hoac mat khau. Thu lai!
Nhap user pass (vi du: admin admin):
```

client 2 đăng nhập sai tài khoản mật khẩu

The screenshot shows a Windows terminal window with three panes. The left pane shows a telnet server running on port 7000. The middle pane shows a client (nc 127.0.0.1 7000) logging in with the username 'admin' and password 'admin'. The right pane shows another client (nc 127.0.0.1 7000) logging in with the username 'guest' and password 'matkha u'. The terminal window has a dark theme and a status bar at the bottom showing the temperature as 27°C and the date as 11/05/2026.

```
ngohoangminh@BaoPC:~/Laptrinhmang$ ./telnet_server
Server is listening on port 7000...

ngohoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
000
Nhap user pass (vi du: admin admin): admin:admin
Sai cu phap! Vui long nhap theo cu phap: user pas
s
Nhap user pass (vi du: admin admin): admin admin
Dang nhap thanh cong!
>

ngohoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
000
Nhap user pass (vi du: admin admin): guest matkha
u
Sai tai khoan hoac mat khau. Thu lai!
Nhap user pass (vi du: admin admin): 
Dang nhap thanh cong!
>
```

client 1 đăng nhập

The screenshot shows a Windows terminal window with three panes. The left pane shows a telnet server running on port 7000. The middle pane shows a client (nc 127.0.0.1 7000) logging in with the username 'admin' and password 'admin'. The right pane shows another client (nc 127.0.0.1 7000) logging in with the username 'guest' and password 'matkha u'. The terminal window has a dark theme and a status bar at the bottom showing the temperature as 27°C and the date as 11/05/2026.

```
ngohoangminh@BaoPC:~/Laptrinhmang$ ./telnet_server
Server is listening on port 7000...

ngohoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
000
Nhap user pass (vi du: admin admin): admin:admin
Sai cu phap! Vui long nhap theo cu phap: user pas
s
Nhap user pass (vi du: admin admin): admin admin
Dang nhap thanh cong!
>

ngohoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
000
Nhap user pass (vi du: admin admin): guest matkha
u
Sai tai khoan hoac mat khau. Thu lai!
Nhap user pass (vi du: admin admin): guest nopass
Dang nhap thanh cong!
>
```

client 2 đăng nhập

```
ngohoangminh@BaoPC:~/Laptrinhmang$ ./telnet_server_r_ver_2
Server is listening on port 7000...
Child process terminated: -1

ngohoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
Nhap user pass (vi du: admin admin): admin:admin
Sai cu phap! Vui long nhap theo cu phap: user pass
Nhap user pass (vi du: admin admin): admin admin
Dang nhap thanh cong!
> dir
chat_server      noblocking_client  sv_server
time_server.c
chat_server.c    noblocking_client.c sv_server.c
topic_server
encode_server    noblocking_server  tcp_client
topic_server.c
encode_server.c  noblocking_server.c tcp_client.c
udp_chat
hello.txt        out.txt            tcp_server
udp_chat.c
http_server      streaming_client    tcp_server.c
udp_chat_ver_2
http_server.c    streaming_client.c  telnet_server
udp_chat_ver_2.c
info_client      streaming_server    telnet_server.c
udp_echo
info_client.c    streaming_server.c  telnet_server_ver_2
udp_echo.c
info_server      sv_client           telnet_server_ver_2.c
udp_sender
info_server.c    sv_client.c         telnetserver_ver_2
udp_sender.c
log.txt          sv_log.txt          time_server
users.txt

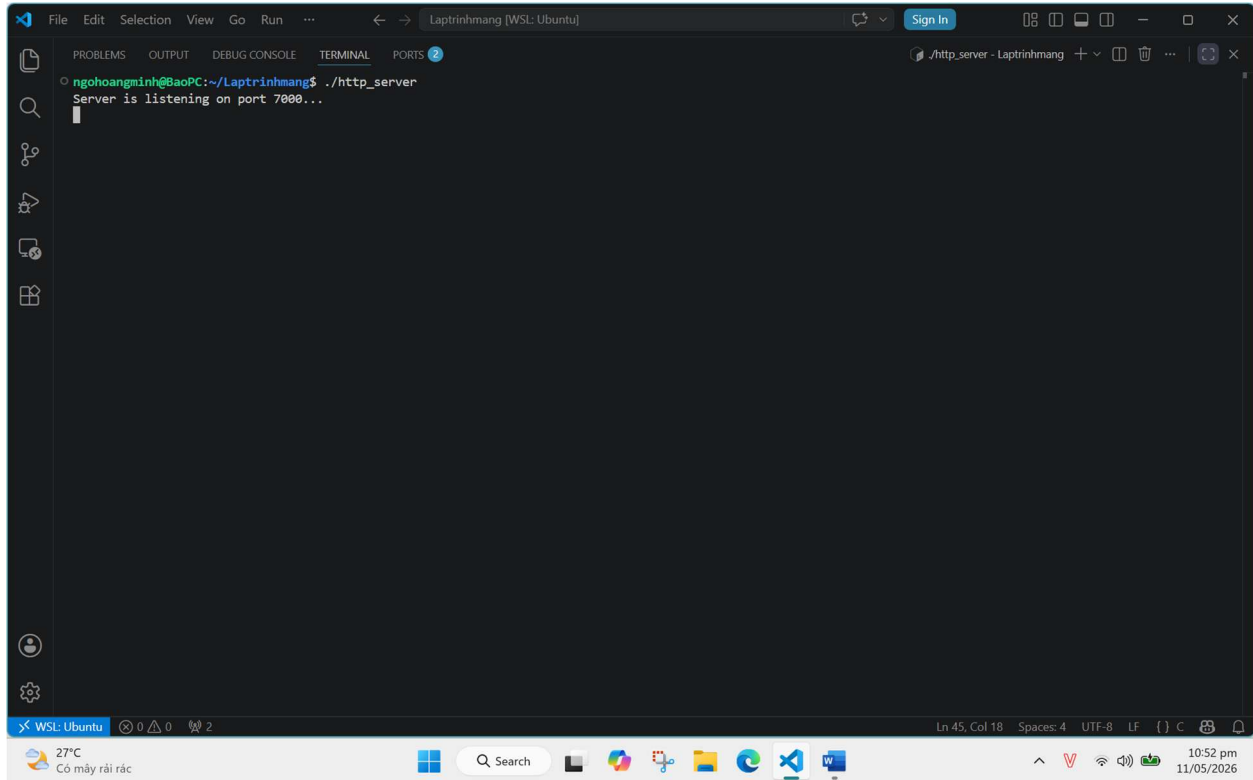
ngohoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
Nhap user pass (vi du: admin admin): guest matkha
u
Sai tai khoan hoặc mat khau. Thu lai!
Nhap user pass (vi du: admin admin): guest nopass
Dang nhap thanh cong!
>
```

client 1 thực hiện lệnh dir

```
1 chat_server noblocking_client sv_server time_server.c
2 chat_server.c noblocking_client.c sv_server.c topic_server
3 encode_server noblocking_server tcp_client topic_server.c
4 encode_server.c noblocking_server.c tcp_client.c udp_chat
5 hello.txt out.txt tcp_server udp_chat.c
6 http_server streaming_client tcp_server.c udp_chat_ver_2
7 http_server.c streaming_client.c telnet_server udp_chat_ver_2.c
8 info_client streaming_server telnet_server.c udp_echo
9 info_client.c streaming_server.c telnet_server_ver_2 udp_echo.c
10 info_server sv_client telnet_server_ver_2.c udp_sender
11 info_server.c sv_client.c telnetserver_ver_2 udp_sender.c
12 log.txt sv_log.txt time_server users.txt
13
```

out.txt sau khi thực hiện chương trình

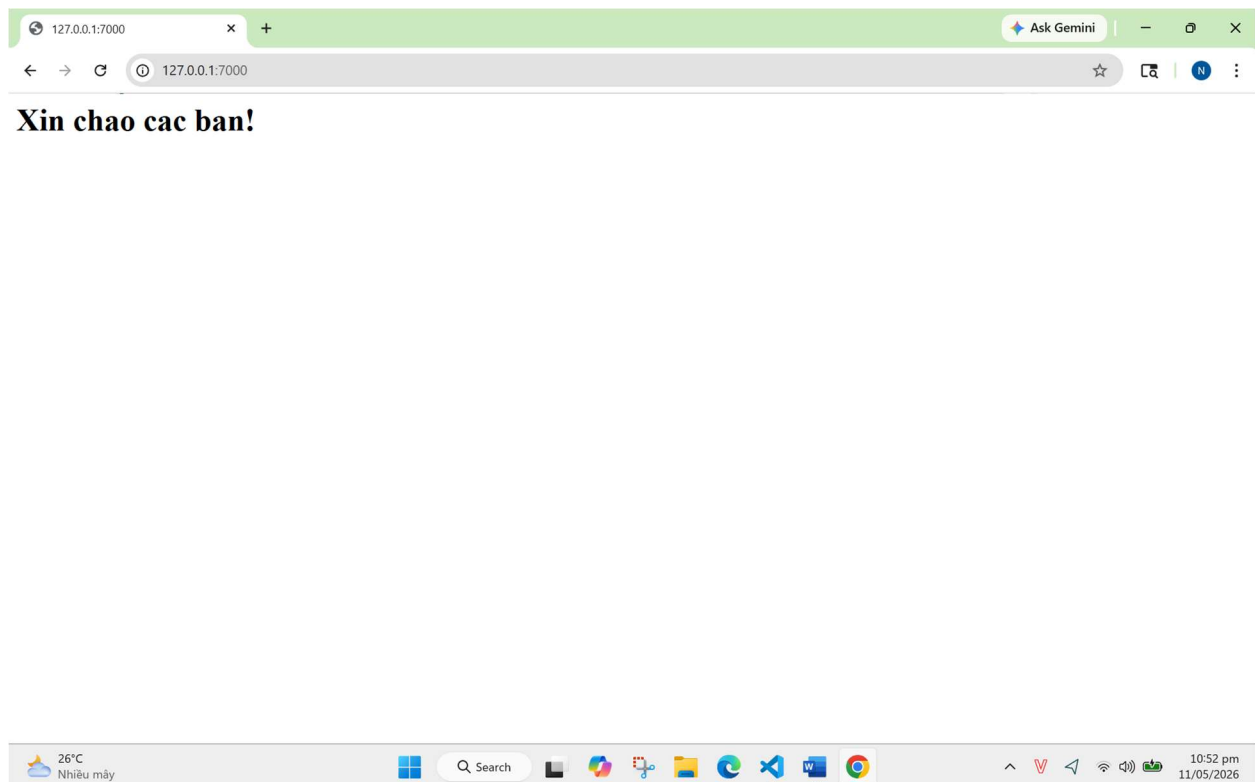
Bài 2: Lập trình lại ứng dụng http_server sử dụng kỹ thuật preforking



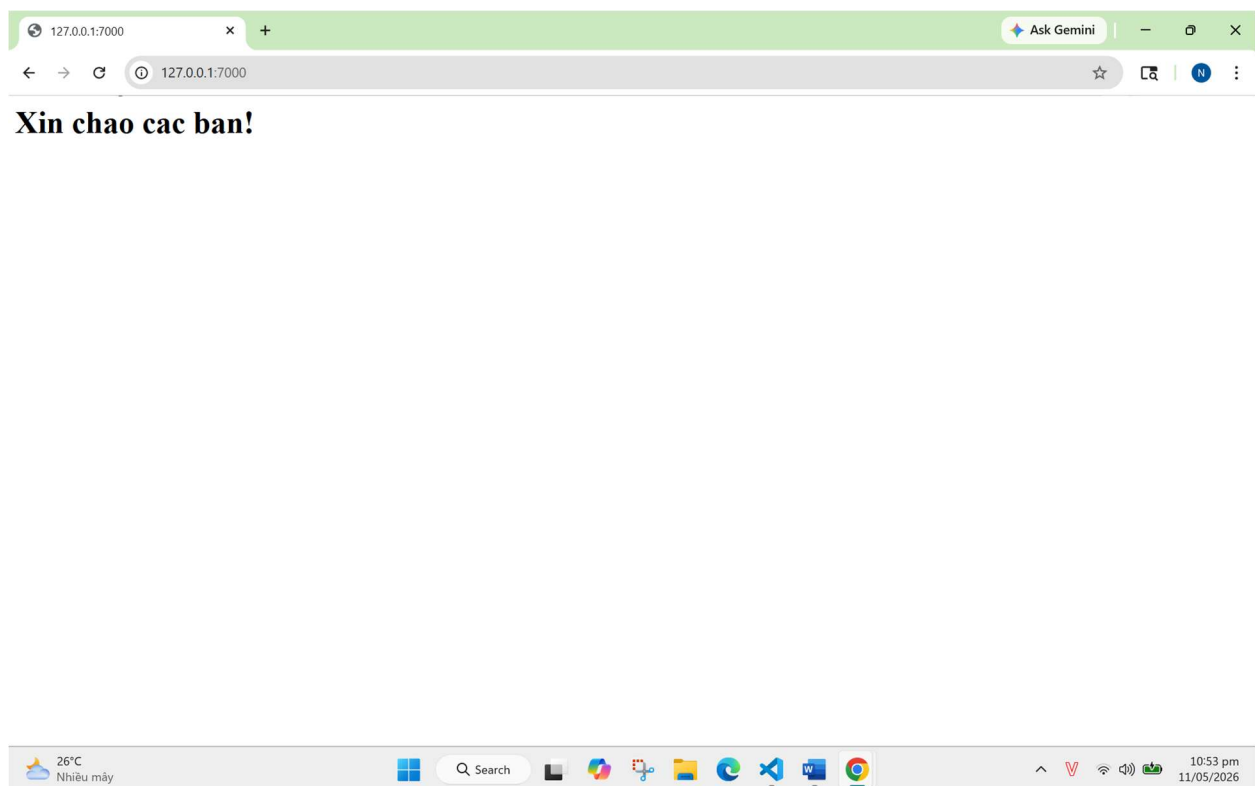
The screenshot shows a Visual Studio Code editor window with a terminal pane open. The terminal is running a command to start an http_server. The output shows the server is listening on port 7000. The terminal window is titled 'Laptrinhmang [WSL: Ubuntu]'. The status bar at the bottom indicates the file is at line 45, column 18, with 4 spaces, UTF-8 encoding, and LF line endings. The system tray at the bottom shows the date and time as 10:52 pm on 11/05/2026.

```
ngohoangminh@BaoPC:~/Laptrinhmang$ ./http_server
Server is listening on port 7000...
```

khởi chạy http_server lắng nghe kết nối socket 7000

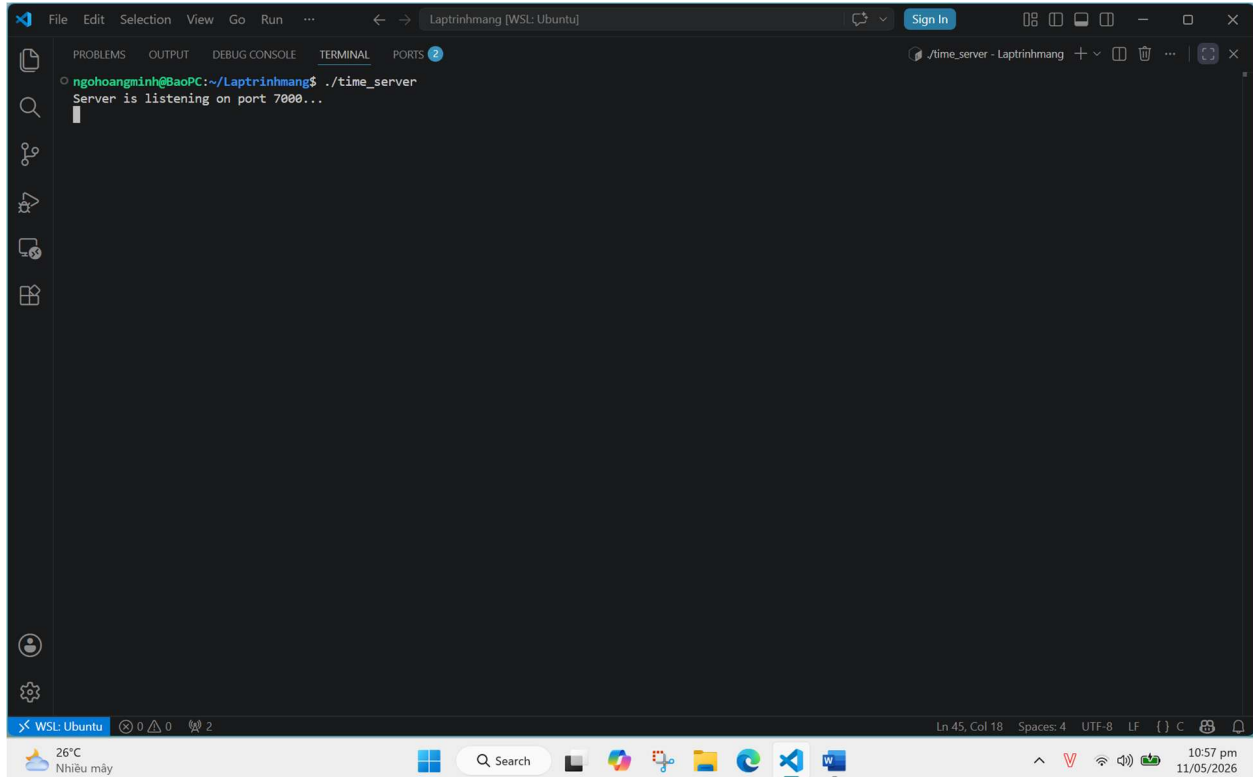


client kết nối



client tiếp theo kết nối

Bài 3: Lập trình ứng dụng `time_server` thực hiện chức năng: Chấp nhận nhiều kết nối từ các client sử dụng kỹ thuật multiprocessing, client gửi lệnh `GET_TIME [format]` để nhận thời gian từ server, format là định dạng thời gian server cần trả về client, các format hỗ trợ: `dd/mm/yyyy`, `dd/mm/yy`, `mm/dd/yyyy`, `mm/dd/yy`, cần kiểm tra tính đúng đắn của lệnh client gửi lên server



The screenshot shows a Windows terminal window titled "Laptrinhmang [WSL: Ubuntu]". The terminal displays the command `ngohoangminh@BaoPC:~/Laptrinhmang$ ./time_server` and the output `Server is listening on port 7000...`. The terminal interface includes a sidebar with icons for file explorer, search, and other tools. The bottom status bar shows the WSL distribution as Ubuntu, the current line and column as Ln 45, Col 18, and the system clock as 10:57 pm on 11/05/2026.

khởi chạy `time_server` lắng nghe kết nối socket 7000

The screenshot shows a VS Code editor window titled "Laptrinhmang [WSL: Ubuntu]". The terminal output is as follows:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
ngoahoangminh@SaoPC:~/Laptrinhmang$ ./time_server
Server is listening on port 7000...
New client connected: 4
ngoahoangminh@SaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
```

The status bar at the bottom indicates "WSL: Ubuntu" and the system temperature is 26°C.

client 1 kết nối tới time_server

The screenshot shows the same VS Code editor window, but now with two terminal panes. The left pane shows the server output, and the right pane shows two client connections:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
ngoahoangminh@SaoPC:~/Laptrinhmang$ ./time_server
Server is listening on port 7000...
New client connected: 4
ngoahoangminh@SaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
000
ngoahoangminh@SaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
000
```

The status bar at the bottom indicates "WSL: Ubuntu" and the system temperature is 26°C.

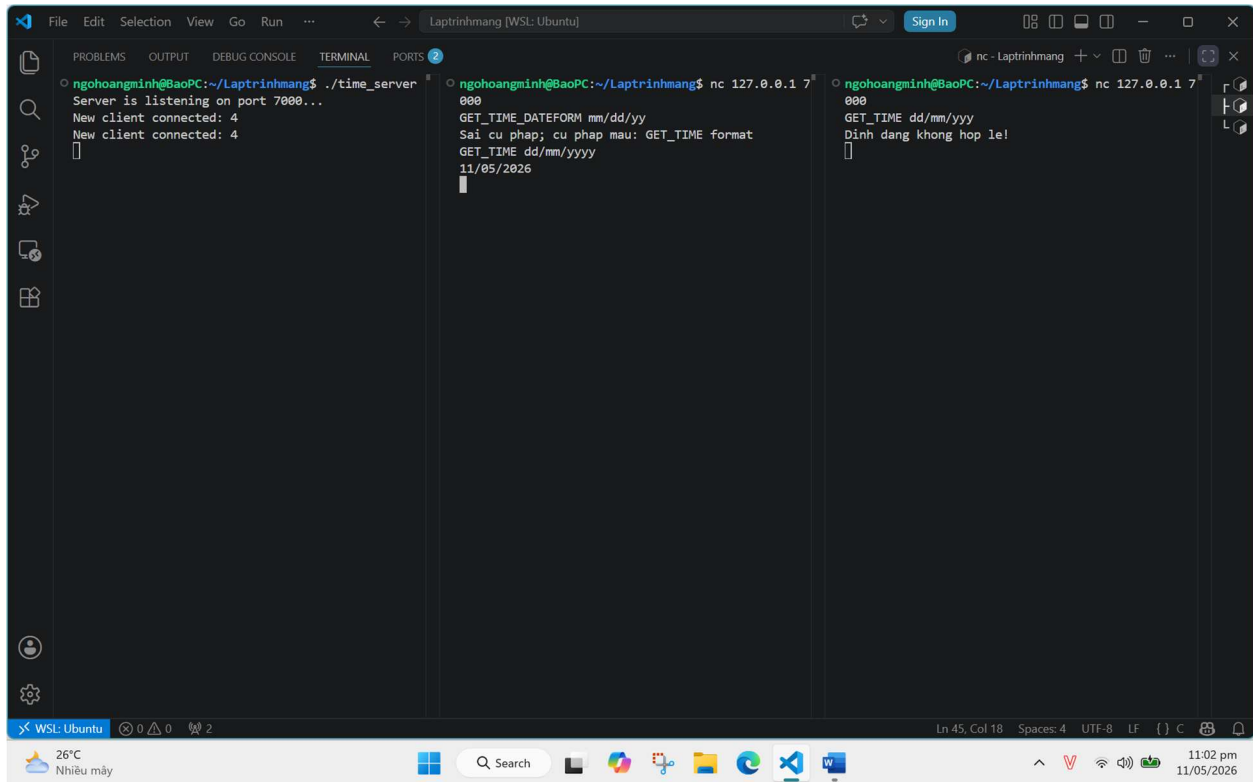
client 2 kết nối tới time_server

```
File Edit Selection View Go Run ... Laptrinhmang [WSL: Ubuntu] Sign In
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
ngoahoangminh@BaoPC:~/Laptrinhmang$ ./time_server
Server is listening on port 7000...
New client connected: 4
New client connected: 4
ngoahoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
GET_TIME_DATEFORM mm/dd/yy
Sai cu phap; cu phap mau: GET_TIME format
ngoahoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
000
```

client 1 sử dụng sai cú pháp

```
File Edit Selection View Go Run ... Laptrinhmang [WSL: Ubuntu] Sign In
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
ngoahoangminh@BaoPC:~/Laptrinhmang$ ./time_server
Server is listening on port 7000...
New client connected: 4
New client connected: 4
ngoahoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
GET_TIME_DATEFORM mm/dd/yy
Sai cu phap; cu phap mau: GET_TIME format
ngoahoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
GET_TIME dd/mm/yyyy
Định dạng không hợp lệ!
```

client 2 dùng định dạng không hợp lệ

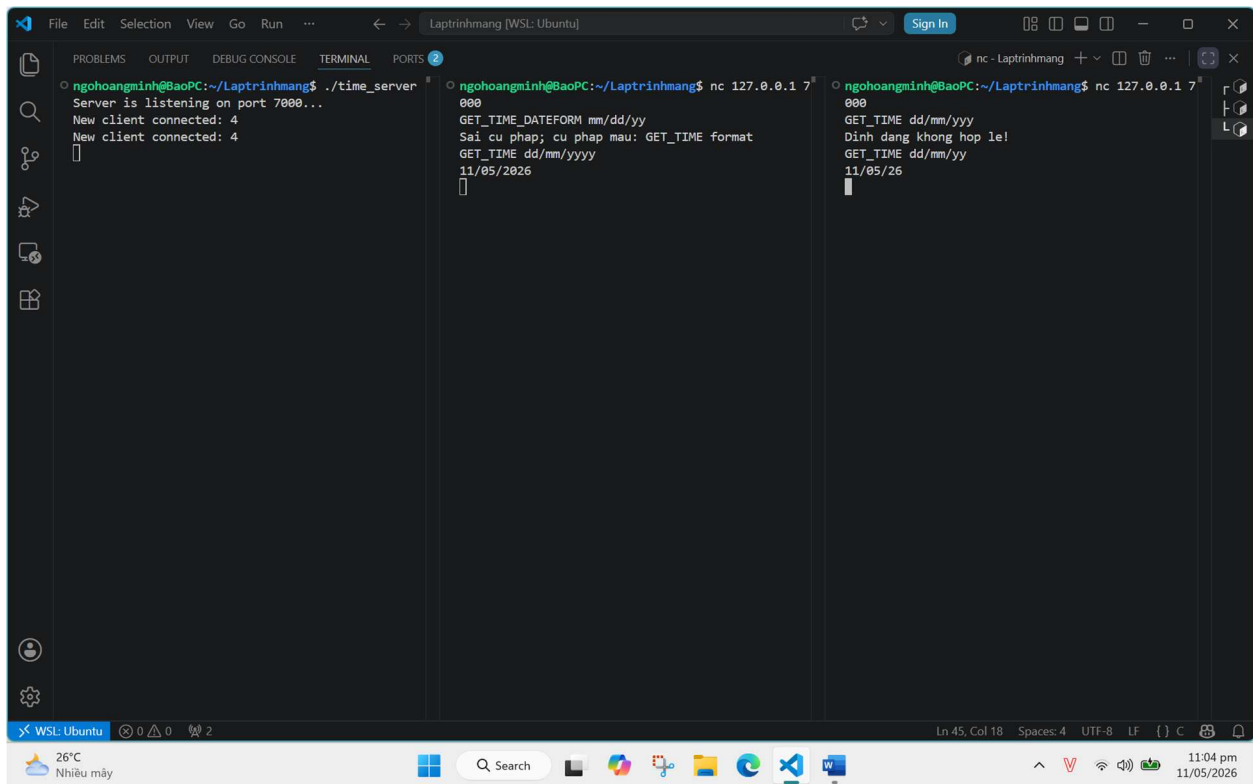


```
ngohoangminh@BaoPC:~/Laptrinhmang$ ./time_server
Server is listening on port 7000...
New client connected: 4
New client connected: 4

ngohoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
000
GET_TIME_DATEFORM mm/dd/yy
Sai cu phap; cu phap mau: GET_TIME format
GET_TIME dd/mm/yyyy
11/05/2026

ngohoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
000
GET_TIME dd/mm/yyyy
Dinh dang khong hop le!
```

client 1 get_time từ server định dạng dd/mm/yyyy



```
ngohoangminh@BaoPC:~/Laptrinhmang$ ./time_server
Server is listening on port 7000...
New client connected: 4
New client connected: 4

ngohoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
000
GET_TIME_DATEFORM mm/dd/yy
Sai cu phap; cu phap mau: GET_TIME format
GET_TIME dd/mm/yyyy
11/05/2026

ngohoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
000
GET_TIME dd/mm/yyyy
Dinh dang khong hop le!
GET_TIME dd/mm/yyyy
11/05/26
```

client 2 get_time từ server định dạng dd/mm/yy

The screenshot shows a WSL terminal window titled "Laptrinhmang [WSL: Ubuntu]". It contains three terminal sessions. The first session on the left is a server running `./time_server`, which outputs "Server is listening on port 7000..." and "New client connected: 4". The middle session is a client running `nc 127.0.0.1 7000`, sending `GET_TIME_DATEFORM mm/dd/yy` and receiving the error "Sai cu phap; cu phap mau: GET_TIME format". The right session is another client running `nc 127.0.0.1 7000`, sending `GET TIME dd/mm/yyy` and receiving the error "Dinh dang khong hop le!". The bottom status bar shows "WSL: Ubuntu", system icons, and the time "11:05 pm 11/05/2026".

```
File Edit Selection View Go Run ... Laptrinhmang [WSL: Ubuntu]
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
ngohoangminh@BaoPC:~/Laptrinhmang$ ./time_server
Server is listening on port 7000...
New client connected: 4
ngohoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
000
GET_TIME_DATEFORM mm/dd/yy
Sai cu phap; cu phap mau: GET_TIME format
GET_TIME dd/mm/yyyy
11/05/2026
GET_TIME mm/dd/yyyy
05/11/2026
ngohoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
000
GET TIME dd/mm/yyy
Dinh dang khong hop le!
GET_TIME dd/mm/yy
11/05/26
WSL: Ubuntu 26°C Nhiều mây 11:05 pm 11/05/2026
```

client 1 get_time định dạng mm/dd/yyyy

This screenshot is identical to the one above, showing the same WSL terminal window with the server and two clients. The middle client session now shows the output of `GET_TIME dd/mm/yyyy` as "11/05/2026". The right client session shows the output of `GET_TIME mm/dd/yy` as "05/11/26". The bottom status bar remains the same.

```
File Edit Selection View Go Run ... Laptrinhmang [WSL: Ubuntu]
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
ngohoangminh@BaoPC:~/Laptrinhmang$ ./time_server
Server is listening on port 7000...
New client connected: 4
ngohoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
000
GET_TIME_DATEFORM mm/dd/yy
Sai cu phap; cu phap mau: GET_TIME format
GET_TIME dd/mm/yyyy
11/05/2026
GET_TIME mm/dd/yyyy
05/11/2026
ngohoangminh@BaoPC:~/Laptrinhmang$ nc 127.0.0.1 7000
000
GET TIME dd/mm/yyy
Dinh dang khong hop le!
GET_TIME dd/mm/yy
11/05/26
GET_TIME mm/dd/yy
05/11/26
WSL: Ubuntu 26°C Nhiều mây 11:05 pm 11/05/2026
```

client 2 get_time định dạng mm/dd/yy